

It's in Your Eyes , Awareness and Control HCI Using EOG Signal

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Introduction

Amyotrophic lateral sclerosis (ALS) this disease makes its patients communicate with the world using eye movement, since their voluntary muscles are paralyzed like physicist "Stephen Hawking". Many efforts are exerted to support this way of communication by tracking or detecting eye movement.

Electro-oculogram (EOG) is an electro-physiological signal generated by eye movement and can be measured using electrodes placed around the eyes and based on Human Machine interface (HMI) we can detect eye movement and patient Interact with machine.

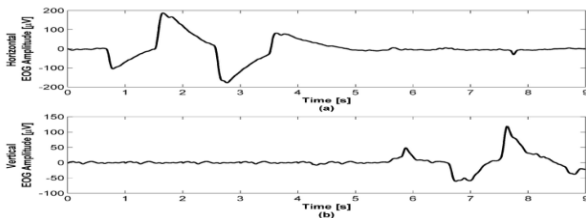


Fig1. Vertical and horizontal EOG Signal

Project Purpose

The objective of this project is to develop a controlling Human-Computer Interface (HCI) system based on electrooculography (EOG).



Methodology

A HCI system is composed of many phases, Signal Pre-Processing, Signal Classification and Computer Interaction.

- EOG will be measured using five electrodes including the ground one, the electrodes will be placed on the forehead. The two channels are arranged vertically and horizontally. Noise and eye blink artifacts will be reduced by filtering using a band pass filter which preserves the EOG spectra
- Finally, classification will be done using Deep Learning Models

Four deep learning models were examined: Convolutional Neural Network (CNN), VGG Network, ResNet Model, and Inception Network.

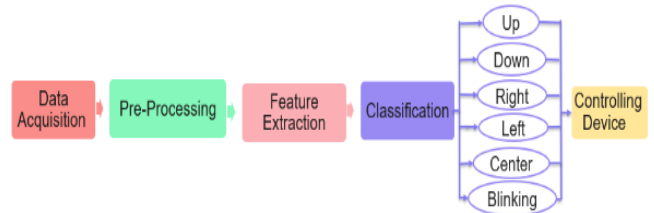
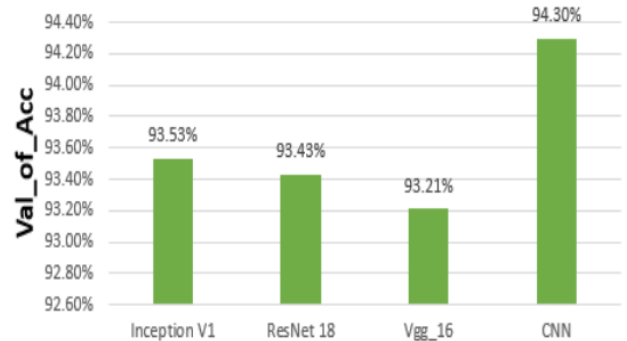


Fig 3.The System Architecture

Results

The experimental results proved the superiority of the Convolutional Neural Network model with average accuracy 94.3, in addition to a small processing time that reached an average of 1 seconds/ action.



Conclusion

The proposed Human-Computer Interface (HCI) system based on electrooculography (EOG) has been achieved with the advanced deep learning models. The experimental result proved that the best average accuracy with CNN model 94.3%.the proposed user interface is usable for patients and help the doctor for examinations. The processing time is 5 sec/action.